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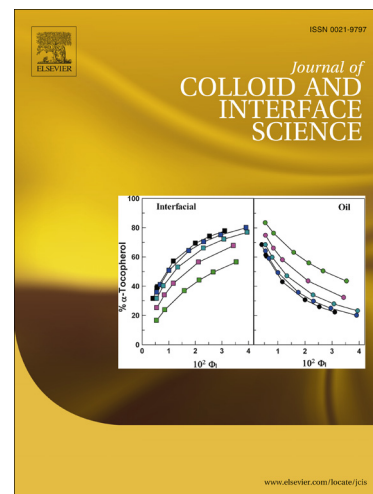
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Effect of vanadium contamination on the framework and micropore structure of ultra stable Y-zeolite

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Abstract

Y-zeolites are the main component of fluid catalytic cracking (FCC) catalyst for conversion of crude petroleum to products of high demand including transportation fuel. We investigated effects of vanadium which is present as one of the impurities in FCC feedstock on the framework and micropore structure of ultra-stable (US) Y-zeolite. The zeolite samples were prepared and characterized using standard techniques including: (1) X-ray diffraction, (2) N₂ adsorption employing non local density functional theory method, NLDFT, (3) Transmittance and Pyridine FTIR, (4) Transmittance electron microscopy (TEM), and (5) ²⁷Al and ²⁹Si MAS-NMR. Results revealed that in the presence of steam, vanadium caused excessive evolution of non inter-crystalline mesopores and structural damage. The evolved mesopore size averaged about 25.0 nm at 0.5 wt.% vanadium loading, far larger than mesopore size in zeolitic materials with improved hydrothermal stability and performance for FCC catalyst. A mechanism of mesopore formation based on accelerated dealumination has been proposed and discussed. Vanadium immobilization experiments conducted to mitigate vanadium migration into the framework clearly showed vanadium is mobile at reaction conditions. From the results, interaction of vanadium with the passivator limits and decreases mobility, and acidity of vanadium into inner cavities of the zeolite capable of causing huge structure breakdown and acid sites destruction. This study therefore deepens insight into the causes of alteration in activity and selectivity of

vanadium contaminated catalyst and hints on a possible mechanism of passivation in vanadium passivated FCC catalyst.

Keywords: Zeolite, Pore structure, Framework, Dealumination, Passivation

1.0 Introduction

Zeolites are an important class of microporous, crystalline aluminosilicates with three-dimensional pore structure widely used in various applications including catalysts [1-14]. Their use as catalysts for different conversion processes has unarguably underpinned the major application in petroleum refining, petrochemical and allied industries. In the case of petroleum refining, zeolite is quite indispensable. The discovery of zeolite catalyst in the 1960s has remained a major turnaround in the petroleum industry that has resulted in improved production and increased profitability [15]. Zeolite is exclusively responsible for cracking catalyst's activity and Y-zeolite, having faujasite structure (Fig. 1), with well-defined pore network ($7.4 \times 7.4 \text{ \AA}$) in its pure crystalline form, is the active component of fluid catalytic cracking (FCC) catalyst and it is used in acidic and rare earth forms to increase the yield of gasoline and diesel fuel from feedstock by cracking heavy paraffins into liquid fuels.

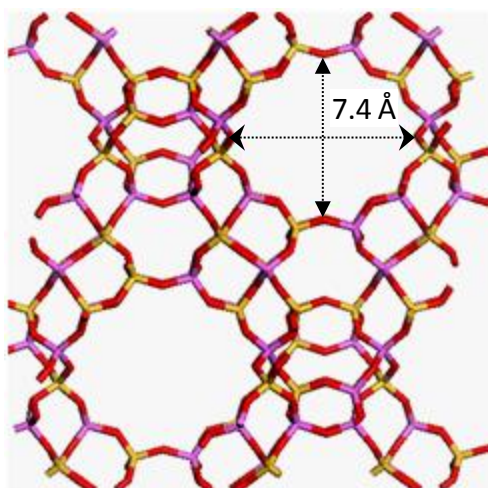


Fig. 1 Structure of Y-zeolite showing average micropore width in the supercage formed by 12-membered ring

During FCC process zeolite catalysts are exposed to certain conditions that can modify important physiochemical properties such as aluminum content and acids strength. Such changes in zeolite composition and properties have significant impacts on catalytic activity, selectivity and stability. It has been established that the harsh FCC unit conditions, together with impurities present in the feedstock are directly linked with properties alterations [16, 17]. Amongst the impurities are metals including sodium [18], iron and calcium [19] and nickel and vanadium [20, 21]. However, nickel and vanadium are grievously detrimental to FCC catalyst's performance [22-24].

Specifically, behaviors of zeolites in the presence of steam and metal contaminants; sodium, vanadium and nickel have been studied. For instance, Pine [25] studied vanadium catalyzed destruction of USY zeolite and results showed steam has ability to destroy the zeolite system; however, his study was on siliceous zeolite. Vanadium and nickel interaction in REY had been reported. The authors discovered that during steam treatment in the presence of vanadium, dealumination of framework alumina and partial collapse of REY structure occur; vanadium also promoted coke formation of the zeolite. However, crystallinity of the zeolite

structure was preserved in the presence of nickel under similar conditions, but coke formation rate was generally on the increase [20, 21]. Tsiatouras and Evmiridis [24], in their study reported a complete structure breakdown of steam treated chromium-exchanged USY zeolite incorporated with vanadium. They authors added structure collapse was nearly proportional to dealumination, and impregnated vanadium selectively removed framework Al atoms. In a study to understand the influence of vanadium on the physiochemical and catalytic properties of USHY zeolite, Torrealba et al. [26] reported that vanadium decreased the stability of the zeolite through decrease in crystallinity and increase of dealumination rate during the hydrothermal treatment, and consequently the activity of the zeolite; but however, this decrease in activity was thought to be due partly to other phenomena, particularly to the exchange of the protonic sites by cationic vanadium species. Generally, vanadium deposited on the surface of the catalyst migrates into the zeolite structure in the presence of steam; however, matters concerning these facts are still debating and no comprehensive report is available in the literature on the action of vanadium on the pore-structure of USY zeolite. In view of the latter, a study of USY zeolite targeting the pore structure in the presence of a contaminating substance such as vanadium becomes necessary to fully comprehend causes of alteration in performance of vanadium contaminated Y-zeolite based catalyst. Results could find application in fabrication of better catalysts and improvement in general FCC catalyst technologies. In this report, we present the effects of vanadium contamination on the framework and micropore structure of USY zeolite using N_2 adsorption by non-local density functional theory (NLDFT) method, powder XRD, MAS-NMR, FTIR and TEM. We have also demonstrated, using physiochemical properties, mitigation of vanadium poisoning of zeolite with a novel synthesized material.

2.0 Experimental methods

2.1 Samples preparation

Zeolite sample of this research, with unit cell size of 24.53 Å, Si/Al ratio 2.8 and sodium content (Na_2O) of 0.42 wt.% was supplied by a catalyst manufacturing Company in Qingdao, China. Working sample was exchanged twice with 1.0 M NH_4NO_3 at 80 °C to remove sodium residue and calcined in air at 500 °C for 4 h. Vanadium USY (xV-USY), where “x” is weight percentage (wt.%) of vanadium in a given sample, were prepared by wet impregnation of vanadyl naphthenate dissolved in n-hexane. Excess organic solvent was evaporated freely in air and oven dried at 120 °C overnight. Oven dried samples were calcined in air at 550 °C for 5 h to obtain the test samples. Unless otherwise stated, all samples were steamed treated for 5 h at a temperature of 760 °C. Passivator as a separate particle was added to oven dried samples by physically admixing desirable amount before subjected to hydrothermal treatment.

2.2 Preparation of vanadium immobilizer

We used yttrium-magnesium oxide (Y/MgO) as our immobilizer here in referred to as “passivator”. Five (5.0) g MgO was dispersed in 40 ml distilled water and stirred for 20 min. Yttrium nitrate ($\text{Y}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$) (4.75 g) dissolved in 10 ml distilled water was added and the mixture allowed stirring for another 30 min. After this time the white suspension formed was transferred to a water bath at 101 °C and allowed to stir until evaporation to dryness. The white solid obtained was oven dried overnight at 110 °C and calcined in air at 600 °C for 5 h.

2.3 Characterizations

Internal structure examination and patterns generation were carried out by powder X-ray diffraction (XRD) of the zeolite samples on a PAN analytical X'pert PRO (Germany), X-ray diffractometer using Cu-K α radiation generated at 40 kV and 40 mA. A scanning range from 5° to 60° was used at a speed of 0.02° 2 θ /s. Crystallinity of the samples was determined using XRD according to standard method in ASTM 3906-03. MAS-NMR was measured for ^{27}Al and ^{29}Si using a Bruker Advance III-400 MHz spectrometer. ^{27}Al was measured at a frequency of 104 MHz at a spinning rate of 12 kHz using a 4-mm probe head, single pulse length of 6 μs ($\pi/20^\circ$) with $\text{AlK}(\text{SO}_4)_2 \cdot \text{H}_2\text{O}$ as standard. ^{29}Si spectra was acquired at 80.4962 MHz at the same spinning rate with reference to DSS-(4,4-dimethyl-4-silapentane-1-sulfonic acid). Transmittance FTIR and acidity characterization were carried out on FTIR spectrophotometer, equipped with a MCT liquid nitrogen cooled detector and a KBr beam splitter. The spectra of the samples were recorded after 64 scans at 4 cm^{-1} resolution. Before analysis all samples were vacuum dried for 2 h to eliminate adsorbed water and gaseous impurities. Samples for acidity studies were sorbed with pyridine for 24 h and the physisorbed species desorbed by vacuum drying at 150 °C for 2 h before analysis. The microstructure and morphology of the materials were examined by transmission electron microscopy, TEM (JEOL JEM-2100UHR) with an accelerating voltage of 200 kV. Textural characterizations including surface area (calculated by BET method, relative pressure (P/P_0) = 0.05–0.28) and pore volume (extrapolated from t-plot, P/P_0 = 0.98) were obtained by nitrogen physisorption method at 77 K following degassing of samples for 6 h at 300 °C, using ASAP 2020 and Tristar 3000 surface area/porosity analyzer both from Micromeritics. Pore size distribution of the catalyst samples were calculated by non-local density functional theory (NLDFT) formalism using non-negative regularization. This method developed by

Lastoskie et al. [27] for modeling slit-like pores on carbons though, was later extended by Saito and Foley [28, 29] to model cylindrical pores in HZSM-5, Y-zeolite and FCC catalyst using N_2 adsorption data and has since gained wide application [30-34]. However, because of very low surface area of 1.0 wt.% V-USY sample, Tristar 3000 was used to measure its textural properties and all of passivated samples.

3.0 Results and discussion

3.1 Effect of vanadium

The zeolite samples including parent-USY and steam-aged (ST-USY), for comparison, were analyzed by powder XRD and results are presented in Fig. 2. Obviously, zeolite peaks of samples containing 0.3 and 0.5 wt.% vanadium decrease in intensity and become broadened, while some are completely lost. On the other hand, sample with 1.0 wt.% vanadium completely lose its zeolite phases.

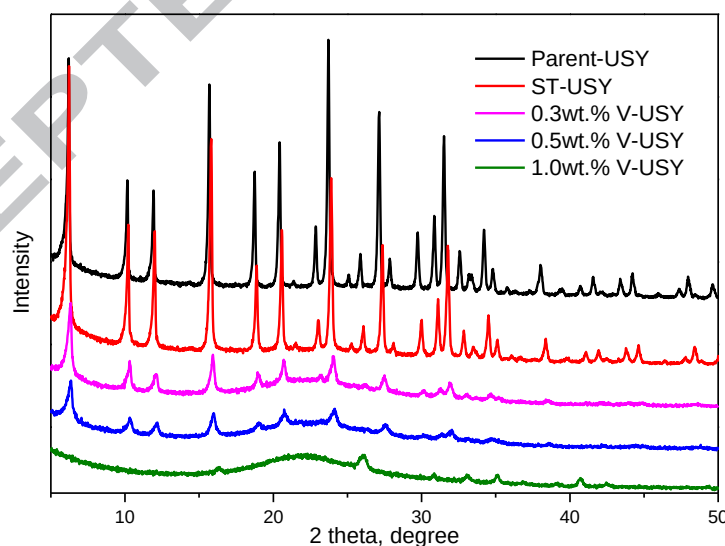


Fig. 2 XRD patterns of zeolite samples showing phase and crystallinity losses

Table 1: Structural properties of zeolite samples

Sample	Si/Al (XRD)	N _{Al} atoms/UC	Structure type
Parent USY	2.82	50.24	Faujasite
ST-USY	4.75	33.38	Faujasite
0.3 wt.% V-USY	242.0	0.79	Faujasite
0.5 wt.% V-USY	*	-2.59	Faujasite
1.0 wt.% V-USY	*	*	Mullite

*not calculated

Table 1 shows properties of the samples obtained from XRD data. Number of aluminum atoms in the unit cell was calculated using Breck-Flanigen's relation [35] (Eq 1).

$$N_{Al/uc} = 115.2 \times (a_o - 24.191) \quad (1)$$

Silicon-to-aluminum ratio, Si/Al, was estimated applying Eq. 1 as follows:

$$Si/Al = \left[\frac{192}{N_{Al}} \right] - 1 \quad (2)$$

The parent-USY has properties of a complete non-denatured USY-zeolite [36, 37]. Aluminum removal from the zeolite framework is observed when sample was subjected to steaming, and on vanadium incorporation and thus it is plausible to say that dealumination is cumulative vanadium-steam induced, with vanadium the more disastrous. Dealumination becomes more severe with concentration of vanadium. As shown in Table 1 loss of Al atoms increases with increase in content of contaminant until absolute structure breakdown and transformation from faujasite to mullite (JCPDS card no: 01-089-2645) when vanadium concentration was 1.0 wt.%. Transformation of the structure of an FCC catalyst containing a REY zeolite, impregnated with 10,000 ppm (1.0 wt.%) vanadium, to mullite has been reported [38].

Textural characterization of materials is one of the most important practices for optimizing their applications. For instance, total surface area and porosity properties are important for catalyst's performance since they determine accessibility of active sites and often relate to catalyst's activity (surface area), while the pore structures including pore size distribution and pore volume control transport phenomena and regulate selectivity in catalytic reactions [39]. The effect of vanadium incorporation on the textural structural properties of the catalyst was investigated by analyzing the data from N₂ adsorption/desorption isotherms. Nitrogen adsorption/desorption isotherms (Fig. 3(I)) were obtained from ASAP 2020. The isotherms are a combination of types (near-type I & IV); type I, containing cylindrical-narrow pores typical of solids having exclusively micropore size with a quantity of mesopores (a) and type (IV), showing bimodal distribution of pores (micro-mesopores, (c–d)) respectively [31, 39]. The figure (Fig. 3(II)) (bottom) shows an entirely destroyed microporous zeolite material with a near parallel (zero) adsorption at $P/P_0 \leq 0.45$. This represents a case of lost micropores leading to cavitations (large mesopores).

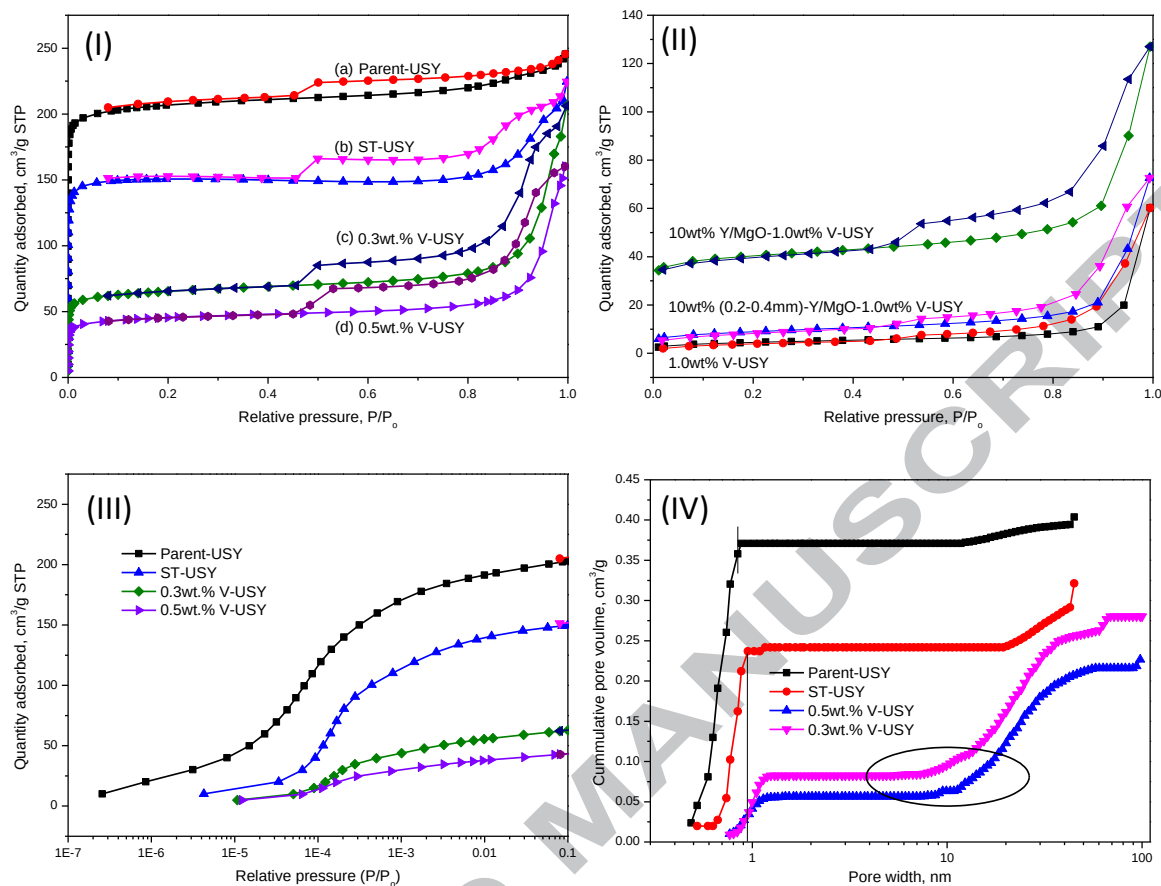


Fig. 3 (I) Nitrogen sorption linear isotherms of the samples; (II) Nitrogen sorption linear isotherm of 1.0 wt.% V-USY and passivated ones; (III) Semilog scale linear isotherms of the samples; (IV) Cumulative pore volume plots obtained by fitting data of DFT model to sorption isotherms.

In the Parent-USY, there was intensive pores filling at relative pressure ≤ 0.1 , corresponding to primary micropores and due to clean surface for physical interaction with the adsorbate molecules; this behavior is also shown by steam-aged sample to an extent. However, samples containing 0.3 and 0.5 wt.% vanadium exhibit different behaviors at the region of relative pressure greater than 0.8, the hysteresis loop of the isotherms show marked phase transition corresponding to different adsorption behavior in the newly formed mesopores. The loops become more convergent in nature ascribed to capillary condensation effect. Also, the reduction in sorption capacity expressed by the samples represents loss of micropores (crystallinity) and

phase transition [40]. As confirmation to formation of larger pores upon vanadium incorporation, semilog scale isotherms are shown (Fig. 3(III)). Apparently, no absorption of gas is feasible at very low pressure region, $1.0 \times 10^{-7} P/P_0$. Generally, below (P/P_0) less than 0.1, absorption was negligible and the isotherm remaining nearly flat, probably due to destruction of the micropore structure of vanadium contaminated samples. Fig. 3(IV) shows plots of cumulative pore volume against pore size, it can be seen that pores filling with diameter (< 1 nm) is dominant in parent and steam-aged samples. However, paradigm shifts appear for contaminated samples, more filling of pores is observed for pores with diameter (> 10 nm) as reflected on the point of inflection and increase in cumulative volume. The slight increase in pores filling pattern of Parent-USY is due to filling of mesopore constituent introduced during initial processing of zeolite Y to USY.

Table 2: Textural properties of zeolite samples

Sample	S_{BET} , m^2/g	^aMPV , cm^3/g	TPV, cm^3/g	DFT report	
				Average MEP width, nm	Average MP width, nm
Parent-USY	625.3	0.2870	0.3797	-	0.73
ST-USY	451.6	0.2265	0.3644	-	0.84
0.3%wt.V-USY	153.3	0.0758	0.3190	23.80	1.00
0.5%wt.V-USY	138.0	0.0514	0.2480	25.38	1.00
*1.0%wt.V-USY	15.74	0.00094	0.0930	-	-

^aDetermined from t-plot; MEP = mesopore; MP = micropore *determined using Tristar 3000.

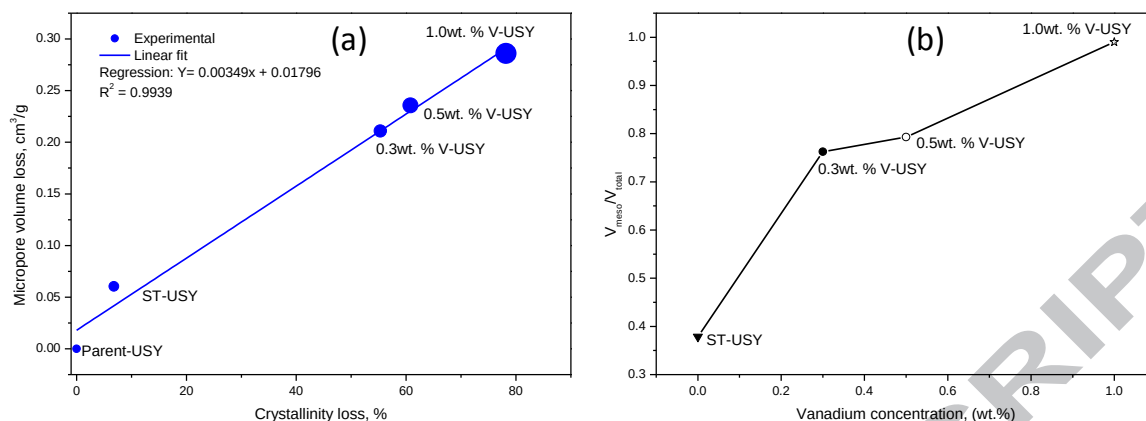


Fig. 4 (a) Correlation of micropores volume with crystallinity (b) Relative mesopore volume ($V_{\text{meso}}/V_{\text{total}}$) of contaminated samples against contaminant concentration after steaming at 760 °C for 5 h.

BET surface area (S_{BET}), total pore volume (TPV) and micropore volume (MPV), and other microporous features of the zeolite samples are summarized in Table 2. The sample containing 0.5 wt.% vanadium shows about 80% BET surface area deterioration with a corresponding percentage reduction of 34.7 and 82.1% in TPV and MPV respectively. In the literature, a near total loss of surface area has been reported for zeolite catalyst containing 1.0 wt.% vanadium loading [20, 41]. Loss of micropore volume of a crystalline solid such as zeolite is a measure of crystallinity loss, indicating structural damage (Fig. 4a). However, there is crystallinity retention when the incorporated impurity was 0.3 wt.% as confirmed in Fig. 2. Diminution in textural parameters for vanadium contaminated samples is highly pronounced when compared with the steam-aged sample. As vanadium loading increased to 1.0 wt.%, structural transformation played out, crystalline zeolite (faujasite) structure was transformed to mullite as evidenced with XRD data. Fig. 4b shows the effect of vanadium incorporation on mesopore formation, which increases with content. This result is no surprise; a near total loss of micropores is shown for 1.0 wt.% V-USY. Because dealumination severity increases with vanadium presence, it would be reasonable to conclude that mesopores formation is caused by defects created as a result of

framework corrosion by accelerated dealumination. Contact of vanadium with zeolite at cracking conditions in the regenerator had earlier been reported to having serious detrimental effects including loss of crystallinity, degradation of surface area, and micropore volume, and dealumination [9, 42, 43].

Pore size distribution analysis was performed using NLDFT formalism from experimental adsorption (N_2 @ -196°C) isotherms data and a model for cylindrical pores in microporous solids (method: non-negative regularization, medium smoothing with $\lambda = 0.01$). Fig. 5 and 6 show the pore size distribution for the samples in full and micropore regions respectively. It can be seen for all samples there are peaks indicating pore size with average width less than or equal 1.0 nm in the micropore region. The observed average pore sizes are slightly higher than constitutional average pore size in conventional USY zeolite except in parent-USY which value is in very close agreement. The average pore width of the parent-USY is 0.73 nm as determined from our experiment, closed to 0.74 nm average pore width of the zeolite with faujasite structure. This value increased to 1.00 nm (DFT model) on incorporation of vanadium into the zeolite due to pore expansion. For average pore width less than 0.74 nm, it is not unusual as such observation had been reported by Saito and Foley [29] in their study to assess the value of the cylindrical pore model for predicting zeolite micro-PSD in Y-zeolites; it is indicative of ultra-micropores. The magnitude of the peaks decreases and become broaden, corresponding to loss of pore volume and variation in pore size (Fig. 6). Nevertheless, another set of pore distribution (average ~ 25.3 nm) emerged at the region of pore width between 10 and 60 nm for vanadium poisoned samples, an indicative of presence of pores of larger size (mesopores). The incremental pore volume plots shown in Fig. 5 are in line with the adsorption isotherms in Fig. 3(I), and with the surface area and pore volume distribution data (Tables 2 and 3). These results reveal that

presence of the metal (vanadium) in the structure of the zeolite causes restructuring of micro-cavity leading to crystallinity losses.

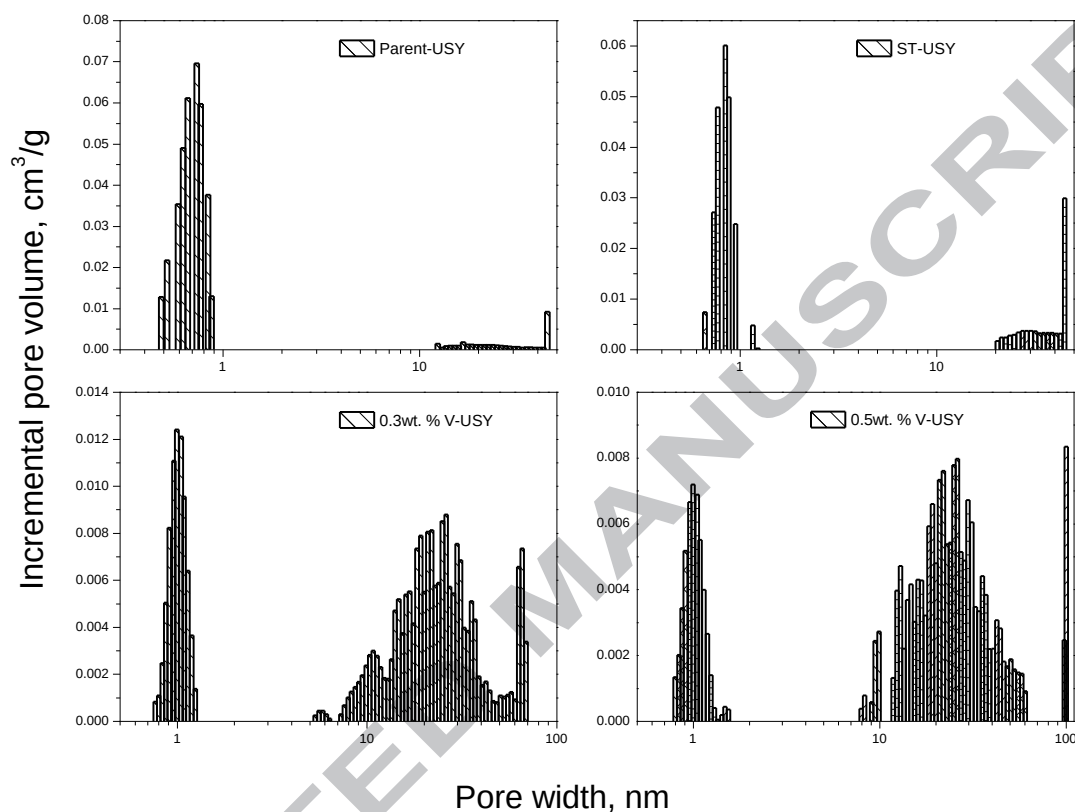


Fig. 5 Pore size distribution calculated using NLDFT method (cylindrical pore model).

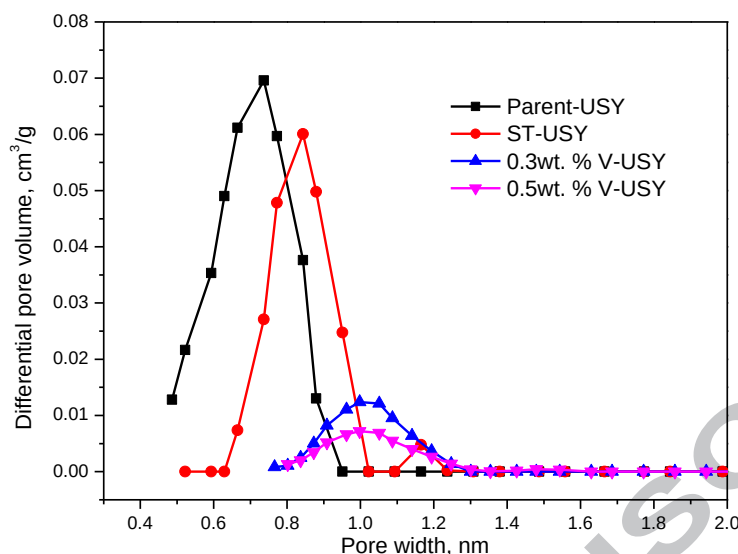


Fig. 6 Micropore region pore size distribution showing average pore width by NLDFT method.

As expected for microporous materials, all samples show peaks near 0.74 nm, typical of Y-zeolitic material. However, this property exhibited for contaminated samples may be due to presence of some portion of micropores but the pores aperture becomes a little wider as shown with DFT pore size distribution model (Fig. 6 and Table 2). Significant pore restructuring is also confirmed with larger pore widths, and differential pore volume decrease marks the quantity of micropores present. To confirm the validity of the model chosen for our analysis, pore volume distribution calculation was taken into account using three different methods as shown in Fig. 7. H-K and DFT models provided best analysis fit, and however, because of the dual porous nature of the samples, the statistical mechanical DFT was selected for further pore volume analysis (Table 3). Obviously, the parent-USY sample contains 82.7% micropore volume. The relatively lower volume may be due to applied treatment for conversion of the zeolite to ultrastable form. The amount decreases to 27.0% when 0.5 wt.% vanadium was incorporated and treated accordingly. Interestingly, as the micropore volume decreased, there was evolution of larger pores comprising low and large mesopores. Low mesopores are “herein” referred to pores with

width ($2.0 \leq \emptyset \leq 20.0$ nm). Samples containing 0.3 and 0.5 wt.% vanadium contain 20.7 and 16.7 % of pores in this range, showing the formation of pores of larger sizes. The size and volume of pores increase with degree of framework crystallinity degradation [44].

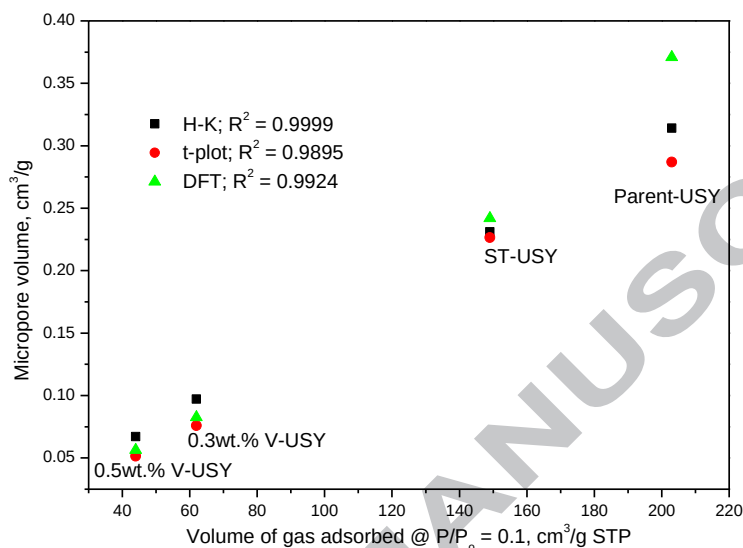


Fig. 7 Comparison of models for pore volume distribution @ P/P_0 0.1 cm^3/g

Table 3: Pore volume distribution

Sample	Pore volume (%)		
	$\emptyset \leq 2.0$	$2.0 \leq \emptyset \leq 50.0$	$50.0 \leq \emptyset$
Parent-USY	82.7	15.8	1.4
ST-USY	66.4	26.9	6.7
0.3wt.% V-USY	30.4	58.3 (*20.7)	11.3
0.5wt.% V-USY	27.0	67.3 (*16.7)	5.7

where $\emptyset$ is pore width, nm, and *% pore volume ($2.0 \leq \emptyset \leq 20.0$)

Characterization of mesoporosity of materials is most frequently accomplished using N_2 physisorption that provides micropore and mesopore volumes and pore size distribution, which the mesopores shape can be ascertained from the shape of physisorption isotherms [33]. However, transmission electron microscopy, in addition to elucidating crystallinity and phases, visualizes mesoporosity within a crystal and the bulk material [45, 46]. TEM images (Fig. 8) reveal clearly excessive holes (mesoporosity) in vanadium treated samples. Also indicated are

crystallinity losses. However, comparing the contaminated samples, it can be observed that crystalline integrity of the sample with higher vanadium concentration is highly impaired. The pore size obtained from TEM is in close agreement with those obtained from physisorption.

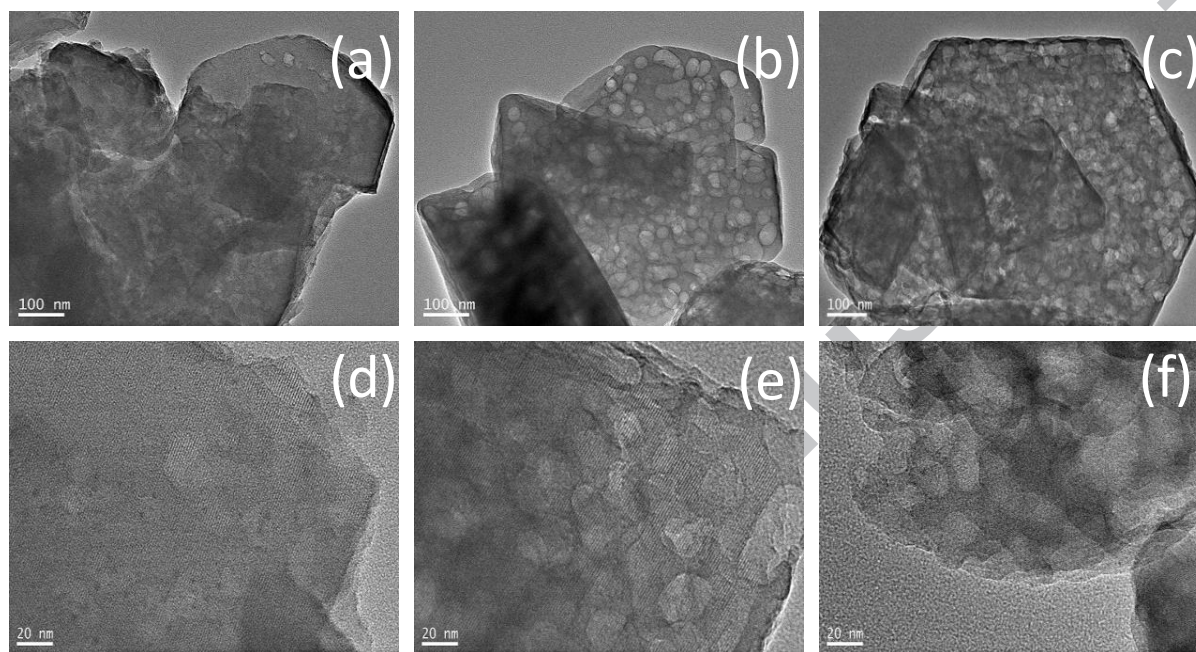


Fig. 8 TEM micrographs of zeolite samples (a) parent-USY; (b) 0.3wt. % V-USY; (c) 0.5wt. % V-USY; (d)-(f) HRTEM of (a), (b) and (c) respectively

Vibrational spectroscopies, such as IR spectroscopy of zeolitic systems provide information about the dynamics of zeolite frameworks and interaction of guest species with catalytically active sites, and IR spectroscopic measurements are widely employed in the studies of zeolitic materials [47]. Fig. 9(I) refers to transmittance IR spectra of our contaminated and passivated zeolite samples and parent-USY. The displayed region furnishes information related to $\nu(\text{T-O})$ vibration consisting both symmetrical ($700\text{-}850\text{ cm}^{-1}$) and asymmetrical ($1000\text{-}1200\text{ cm}^{-1}$), and aluminosilicate tetrahedral bending vibrations [48]. As can be seen from the figure, the structural disturbance is reflected by marked changes in the intensity of the bands which is indicative of structural/alteration damage. Highly perturbed structures show shift in vibration

band to higher frequencies due to structural reconstruction of the framework constitutional atoms caused by cumulative hydrothermal treatment and vanadium interaction. A typical Y-zeolite band ca. 1048 cm^{-1} is visible in parent-USY, however, this band shifts to higher frequency in contaminated samples, due to dealumination of tetrahedral framework aluminum atom [49], giving rise to a characteristic of high silica zeolite [50]. Dealumination has been reported to result in shrinkage of total zeolite framework because of Al-O and Si-O bonds differences [40]. Additionally, $\nu(\text{T-O})$ bending vibrations at near 460 cm^{-1} become more intense, an observation Sohn et al. [49] referred to as “orderliness within the framework caused as a result of dealumination by steaming”, but herein however, we refer to as “disorderliness or defect” in the framework caused by accelerated dealumination as revealed by XRD and TEM data. In 1.0 wt.% V-USY sample, there is complete loss of band near 1200 cm^{-1} present in parent-USY sample and other less contaminated samples, marking Al-O bond breaking. Also, a network of bands in the symmetrical region is redefined. Fig. 10 shows the effect of vanadium on the framework of the zeolite samples.

Adsorption of suitable probe molecules such as pyridine can reveal the amount of acid/basic sites present in a solid catalyst. In this study, the concentrations of Brönsted acid as PyH^+ and Lewis acid as PyL were determined by DRIFT spectroscopy after adsorption of pyridine and desorption at $150\text{ }^\circ\text{C}$, in the $1650\text{--}1400\text{ cm}^{-1}$ region of the IR spectra (Fig. 9(II&III)). The bands corresponding to Brönsted and Lewis acid sites at 1545 and 1445 cm^{-1} respectively, are shown. Bands intensities vary with acid sites concentration in the samples. As can be clearly seen, samples with vanadium content diminish intensities as vanadium loading increased, related to decrease density and/or destruction of acids sites. Additional bands at 1490 cm^{-1} and near

1595 cm^{-1} can be ascribed to pyridine molecules chemisorbed on mixed acid sites and on a second site of Lewis acid, respectively [51].

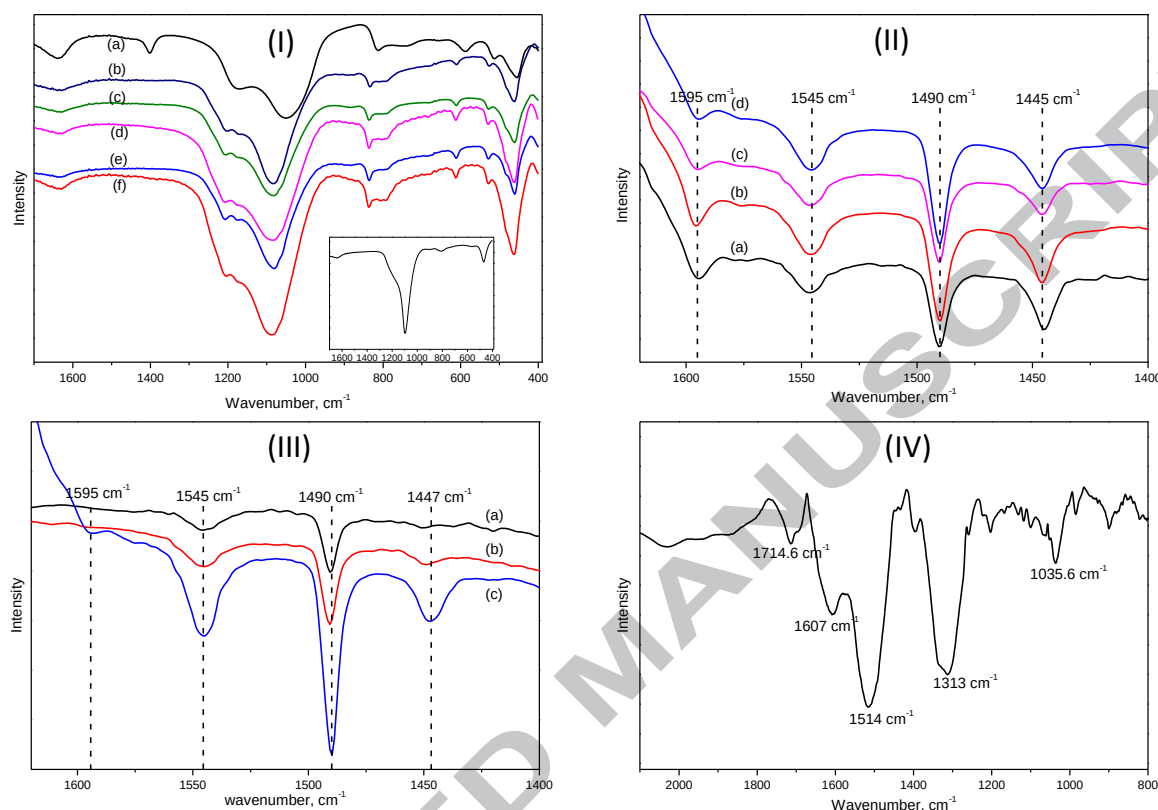


Fig. 9 (I) Transmittance FTIR spectroscopy of zeolite samples showing symmetrical and asymmetrical $\nu(\text{T-O})$ vibration for (a) Parent-USY; (b) 10 wt.% Y/MgO-1.0 wt.% V-USY; (c) 10 wt.% Y/MgO-0.5 wt.% V-USY; (d) 0.3 wt.% V-USY; (e) 10 wt.% Y/MgO-0.3 wt.% V-USY; (f) 0.5 wt.% V-USY, insert is 1.0 wt.% V-USY, (II) FT-IR spectra of PyH^+ and PyL bands of the zeolite samples desorbed at 150 $^{\circ}\text{C}$ for 2 h after adsorption for 24 h (a) 0.3 wt.% V-USY; (b) 10 wt.% Y/MgO-0.3 wt.% V-USY; (c) 0.5 wt.% V-USY; (d) 10 wt.% Y/MgO-0.5 wt.% V-USY. (III) FT-IR spectra of PyH^+ and PyL bands of the zeolite samples desorbed at 150 $^{\circ}\text{C}$ for 2 h after adsorption for 24 h. (a) 1.0 wt.% V-USY (b) 10 wt.% (0.2-0.4mm)-Y/MgO-1.0 wt.% V-USY (c) 10 wt.% Y/MgO-1.0 wt.% V-USY (V) Acid sites probe of passivator before addition to vanadium impregnated catalyst.

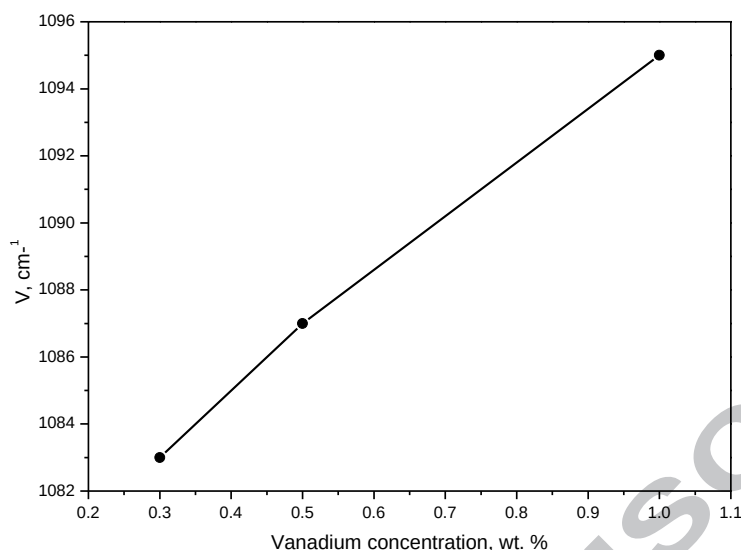


Fig. 10 Effect of vanadium on zeolite framework

Dealumination of zeolite framework suggested by IR, TEM and XRD results was further investigated by MAS-NMR. Fig. 11 (I) shows ^{29}Si MAS-NMR of the samples and parent-USY. As labeled, parent and steam-aged samples exhibit native characteristics of such zeolite samples [52, 53]. However, to confirm migration and probe when severe impairment of zeolite structure/dealumination by vanadium occurs, a sample impregnated with vanadium and calcined (but without steam treatment) was studied, as can be seen, pattern similar to the parent sample is obvious. Vanadium contaminated samples display totally different patterns, an indicative of rearrangement of framework composition atoms. In the steam-aged sample, a siliceous zeolite (Si, 0Al) resonance near -107 ppm is prominent whereas for contaminated samples, merging of aluminum environment peaks, which broadened and shift upfield, is evident. New peaks such as the one at 109.5 ppm develop which Lutz [40] described as disturbance in framework structure and indicate presence of siliceous extra-framework species. It is generated by crystallographically non-equivalent tetrahedral centres arising from silanol group on the zeolite

surface. Comparing the characteristic spectra of the treated samples with the parent-USY reveals zeolite has undergone extensive dealumination [53].

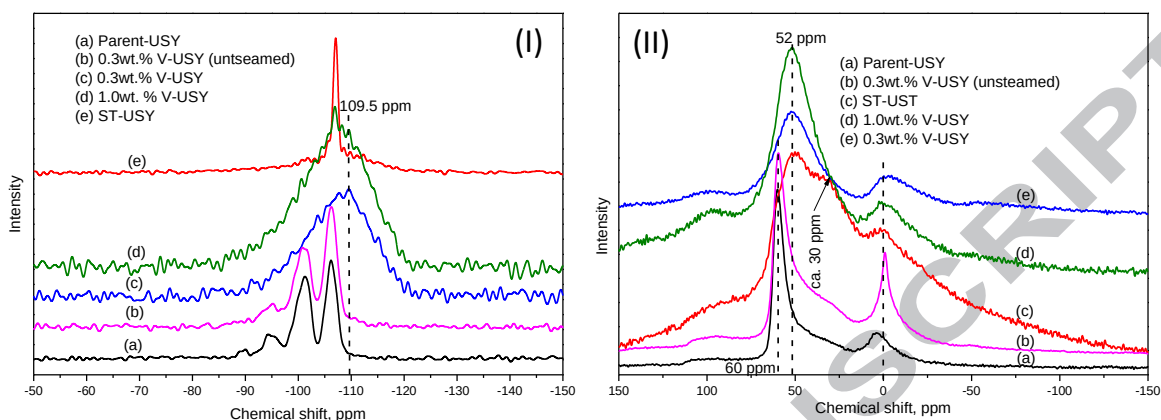
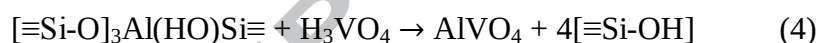


Fig. 11 (I) ^{29}Si MAS-MNR of catalyst samples (II) ^{27}Al MAS-MNR of catalyst samples.

From ^{27}Al MAS-NMR spectra, resonances near 0 and 60 ppm are attributed to Al (VI) and Al (IV) entities, corresponding to non-framework and framework aluminum species respectively. As shown in Fig. 11(II), parent-USY and vanadium impregnated and calcined (but unsteamed) USY exhibit spectral features of structurally unimpaired zeolite. This implies steam aids vanadium destructive action. However, the increased peak at 0 ppm for unsteamed sample is an indication of presence of EFAL. Obviously, alteration of aluminum micro-environments is due to combination of treatments, simulating conditions in a typical FCC unit which the samples were subjected to. Excessive dealumination of framework lattice is evidenced by fully coordinated non-lattice center at 0 ppm, compared with parent-USY at about 4 ppm. This is sequel to framework aluminum atoms losses, indicated by upfield peaks shift to about 52 ppm and suggesting the presence of a different aluminum environment. However, investigation of this is beyond the scope of the present work. Furthermore, the resonance ca. 60 ppm becomes broadened and reduces intensity, an indication of structure disorder. In addition a new peak developed in the steam-aged sample near 30 ppm, is attributed to Al (V) or to distorted Al (VI/IV)

environments [52, 54]. It has been shown that by comparing Al MAS-NMR with Al CP MAS-NMR, steam-aged HY has a substantial portion of the resonance attributed to Al (IV) results from the presence of extraframework Al species [55]. Therefore, it is noteworthy to mention the prominence of Al (IV), framework aluminum centre in vanadium contaminated samples may be due to transformation of crystalline aluminosilicate structure to another structure such as mullite material which also present element of framework aluminum atoms.

Based on our experimental results, we have proposed a mechanism for the formation of mesopores in vanadium contaminated zeolite. USY zeolites are designed to be thermally stable. In the presence of steam and some foreign substances the original stability is compromised. Steaming alone extracts few aluminum atoms from the zeolite framework and creates limited vacancies [56, 57]. Portion of the vacancies (cavitations) are healed by silicon mobile species, resulting in the formation of mesopores [58]. However, in the presence of vanadium, excessive/accelerated dealumination of framework aluminum atoms occurs requiring proportionate filling by silicon atoms. The detail is summarized in the equations below:



Under FCC regeneration conditions vanadium, which has been reported to exist in (+5) oxidation state [22, 25, 59] (V_2O_5), undergoes reaction to form mobile vanadic acid (H_3VO_4) on the surface of the zeolite. Vanadic acid formed undergoes intra-particle migration to the zeolite framework where it attacks and destroys strong acid sites, $\text{Si}(\text{OH})\text{Al}$, via extraction of tetrahedral aluminum atoms in the zeolite crystal, causing it to collapse, and forms solid aluminum vandate (AlVO_4) solution (Eq. 4). Due to unstable nature of AlVO_4 under FCC conditions, it decomposes and

H_3VO_4 acid is re-formed to continue aluminum extraction (dealumination) process in a catalytic fashion (Eq. 5). According to this mechanism, vanadic acid would function as a catalyst since it is not consumed [17]. From the perception of the above mechanism and our XRD data (N_{Al}), excessive dealumination accompanying vanadium contamination leaves much vacancies/defects (silanol nests) which are insufficiently healed by unstable mobile silicon species (dealumination >>> silicon substitution). As a result, cavitations formed from expelled aluminum and silicon mobile species are depicted as mesopores revealed from our TEM data. It is worth noting that the formed mesopores are non-intercrystalline, most of which are spherical, and some few which coalesce to larger pores through micropore openings having variant shapes. The framework dislodgement of aluminum atoms are depicted in Fig. 12 and scheme 1.

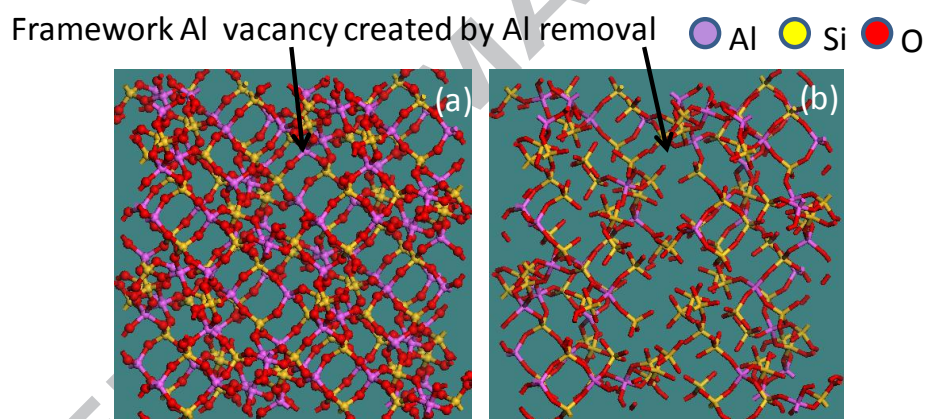
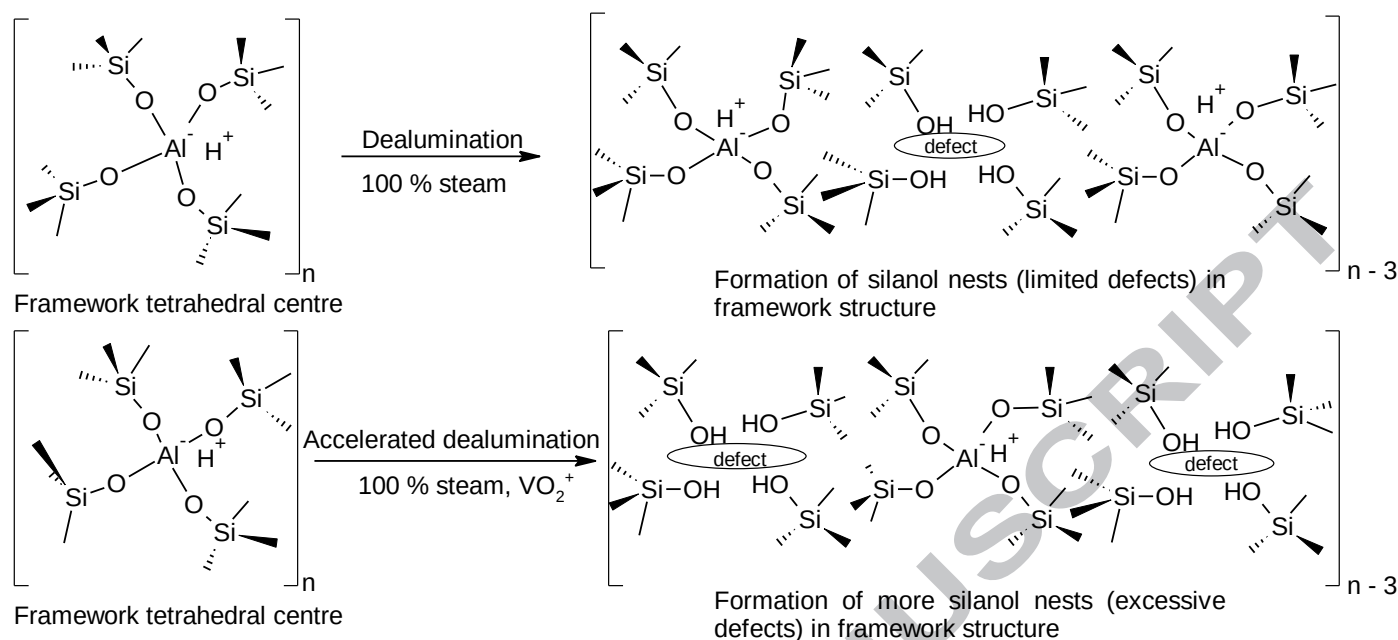


Fig. 12 3D atomistic model of faujasite structure zeolite (a) with full framework aluminum atoms (b) with partial framework aluminum atoms



Scheme 1: Formation of defect/vacancy in framework structure of zeolite by vanadium

3.2 Vanadium passivation

In the light of the foregoing results and the literature, experiments were conducted to mitigate migration of vanadium into in the framework of zeolite. Table 4 shows textural results of samples containing passivator. Comparing data in Table 4 with those of Table 1, it is very obvious there are remarkable changes in the properties of contaminated samples containing additive. This test is evaluated essentially of micropore area and volume, and compared to the samples without additive. In the sample containing additive, (60 μm particle size, 10 wt.% Y/MgO) on 1.0 wt.% V-USY, total and micropore surface areas (calculated by BET method) of 125.7 and 88.9 m^2/g have been obtained against 15.7 and 1.94 m^2/g respectively. Micropore volume drastically increased from 0.00094 to 0.0462 cm^3/g , it signifies reduced destruction by vanadium. In our vanadium immobilized samples, improved FTIR spectral patterns are also presented (see Fig. 9(I)). Specifically, in 1.0 wt.% V-USY sample the peak at near 1200 cm^{-1} and other symmetrical peaks which had lost due to dealumination ($700\text{--}850\text{ cm}^{-1}$) re-appeared. This

is indicative of minimum vanadium destructive action due to addition of passivator which reduced migration of vanadium into the framework of the zeolite sample.

Table 4: Phase retention in passivator containing zeolite samples

Sample	S_{BET} , m^2/g	MPA, m^2/g	MPV, cm^3/g
Y/MgO	55.1	0.71	-
5wt.% 60 μm	30.2	8.1	0.0041
Y/MgO/1.0wt.% V-USY			
10wt.% 60 μm	202.3	176.7	0.0912
Y/MgO/0.3wt.% V-USY			
10wt.% 60 μm	163.8	123.1	0.0637
Y/MgO/0.5wt.% V-USY			
10wt.% 60 μm	125.7	88.9	0.0462
Y/MgO/1.0wt.% V-USY			
50wt.%/0.2-0.4mm/Y/MgO/	28.4	6.3	0.0032
1.0wt.% V-USY			

Properties measured with Tristar 3000.

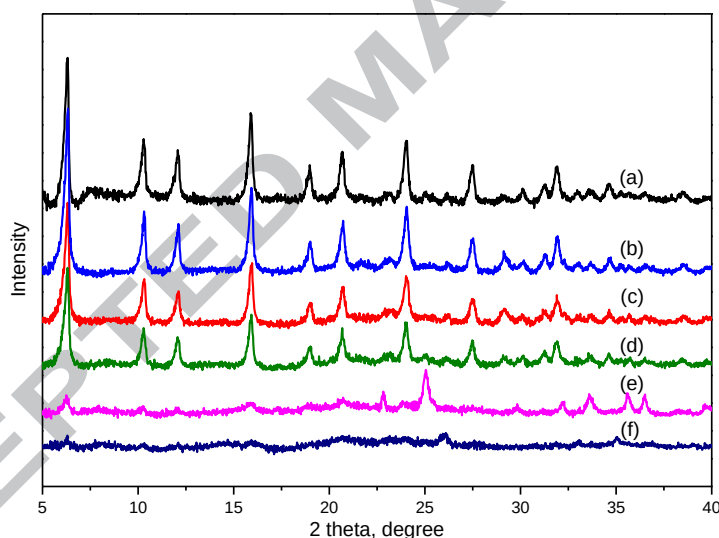


Fig. 13 XRD pattern of vanadium passivated zeolite samples showing phase retention (a) 5 wt.% Y/MgO-0.5 wt.% V-USY (b) 10 wt.%Y/MgO-0.3 wt.% V-USY (c) 10 wt.% Y/MgO-0.5 wt.% V-USY (d) 10 wt.% Y/MgO-1.0 wt.% V-USY (e) 5 wt.% Y/MgO-1.0 wt.% V-USY (f) 50 wt.% (0.2-0.4mm) Y/MgO-1.0 wt.% V-USY

Acidity study was also conducted for the samples with passivator and results are shown (Fig. 9(II & III)). To show that the passivator does not have acid sites capable of cracking, we have presented a spectrum of pyridine sorped passivator (Fig. 9(IV)). Obviously, samples containing passivator retained acidity compared with the samples without passivator. These

results prove ability of the passivator to prevent migration of vanadium into the internal structure of zeolite where acid sites are located [60]. It is important to note here that, a 10 wt.% passivator is probably excessive for vanadium concentrations lower than 1.0 wt.%, as confirmed in acid concentration of passivated samples (Fig. 9(III)).

4.0 Conclusion

Vanadium in the petroleum feedstock affects catalyst in several ways including loss of micropores, occupies the inner pores of zeolite, acid sites losses, accelerated dealumination leading to the formation of larger mesopores, and destroys the framework structure. Its contamination of a catalyst produced a continuous loss of textural properties, as a function of the quantity of the metal. At vanadium contents above catalyst tolerable level in petroleum feedstocks, excessive loss of crystallinity (microporosity) and introduction of large mesoporosity would result in collapse of zeolite structure. The instability of vanadium poisoned zeolite catalyst at FCC regeneration conditions is related not only to microporosity and framework degradations, but also to the formation of a material with different structure which lacks suitable acid sites and catalytic cracking properties. Vanadium immobilization experiment showed that mitigation of vanadium destructive effects preserved important catalytic properties such as acidity, crystallinity and surface area.

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Graphical Abstract

